

Mobile phones equipped with contactless smart cards employ near-field communication (NFC) technology to exchange data between the mobile device and nearby smart card readers. The technology combines the smart card interface as well as the reader interface in the mobile device so that the mobile device can communicate with card readers and other NFC devices/mobile phones.

At the time of making payments, the mobile phone user waves his mobile phone (equipped with contactless smart card) near a reader module installed inside the store or in a public transport system. In order to make the transaction more secure, a PIN is required for authentication, which is automatically supplied by the smart card. After the authentication is verified, the transaction completes and the payment is automatically deducted from the prepaid account of the mobile

technology. It has a URL and displays descriptions of various products that the user may intend to buy. The mobile online payment mechanism combines direct debiting systems of some partner banks with SMS payment confirmation through the mobile phones.

It works like this: The customer selects a product from the webpage displayed in his mobile phone screen and places order for the product through the website. Next, he types the mobile phone number and the password to make the payment. The partner bank verifies the password and, if found okay, sends a text message to the customer. Next, the customer sends an SMS to confirm the payment and the payment is directly debited from the customer's bank account.

The entire payment process is simple, quick and user-friendly as there is similarity to the familiar set of online payment pages. This plug-and-play

through his mobile phone, the user types the account number and sends to the issuing bank through SMS. The mobile wallet's automated system checks for sufficient balance in the user's mobile wallet account and the payment amount is automatically debited from the user's account to perform the required purchase transaction. All the payment requests to the participating banks must be initiated from the registered mobile phone number of the mobile user. After successful completion of the payment transaction, both the merchant (supplier of the product) and the mobile user get notification about payment confirmation from the mobile-wallet issuing bank.

Two different mobile wallet account holders can also transfer money from one mobile wallet account to another. In order to make the transaction safe and secure, proper encryption and authentication techniques are employed.

Mobile wallets have become popular in developing countries such as Kenya and the Philippines, where people at the bottom-line of the pyramid are usually underserved by traditional banks and lenders. The major players in mobile wallets are Globe's G-Cash in the Philippines, PLDT's Smart Money and mPesa in Kenya.

Services in operation world over

NTT DoCoMo. NTT DoCoMo is the prominent mobile phone operator in Japan that provides mobile phone, video phone, Internet phone and mail (Internet mail and SMS) services to the customers. It has established a simple and effective mobile payment solution in collaboration with Sony.

DoCoMo mobile phones are embedded with Sony Felica contactless chips that enable mobile payments in shops fitted with a Felica smart card reader. These Felica-powered smart card handsets can be used for a variety of functions such as debit card transactions and credit card transactions, and also act as personal identification card.

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user or directly charged to the bank account of the user.

NFC-based contactless mobile payment finds wide applications in transportation services, toll-tax collection, transit fare collection in mass-transit networks, parking fee collection, vending machine collections and other unattended POS terminals where the users can pay with their smart card enabled mobile phones sitting inside their car while driving. Leading players in this field are MobiPay, MobiPurse, MobiCard, NTT DoCoMo and FeliCa (Sony).

Mobile Web payments

In this mode of mobile payment, the payment is made through webpages displayed in the micro-browser of the mobile phone. The webpage is displayed following wireless application protocol (WAP) and associated

payment system offers a smooth and transparent micro payment solution to mobile users and is gradually emerging as a potential payment mechanism in mobile commerce. However, special care should be taken to secure such Web payments and proper password authentication procedures should be implemented.

Vodafone has recently launched a mobile Web payment system called mpass that is accessible to each and every mobile phone user in Germany.

Mobile wallets

In mobile wallets system, the mobile phone user opens a mobile wallet account in an issuing bank and deposits cash that will be used in future mobile payments. The user gets the mobile wallet account number from the issuing bank.

At the time of making any purchase